InvadeAI: INteractive ADvertising with Multimodal AI

Andrew Dunton†
Department of Applied Data Science
San Jose State University
San Jose, CA
andrew.dunton@sjsu.edu

Riddhi Vyas†
Department of Applied Data Science
San Jose State University
San Jose, CA
riddhikamleshkumar.vyas@sjsu.edu

Harshal Shah†
Department of Applied Data Science
San Jose State University
San Jose, CA
harshaljaydeep.shah@sjsu.edu

Soujanya Sankathala†
Department of Applied Data Science
San Jose State University
San Jose, CA
soujanya.sankathala@sjsu.edu

Mohammad Masum*
Department of Applied Data Science
San Jose State University
San Jose, CA
mohammad.masum@sjsu.edu

***Abstract*— In this paper, we introduce INvadeAI, a novel interactive advertising framework that leverages multimodal AI to monetize Large Language Models (LLMs) while enhancing digital advertising experiences. Our multi-stage framework integrates YOLOv8 for product detection in visual media, BLIP3 and OCR for caption generation and text extraction, and multimodal LLMs for product link retrieval. Unified within an interactive user interface, INvadeAI enables seamless product discovery and purchasing, creating a potential new revenue stream for computationally intensive AI systems. We applied a fine-tuned YOLOv8 model trained on 25 custom product classes—achieving mAP@50 of 0.535—alongside the base YOLOv8 model's 64 product classes, enabling detection of products across 89 consumer categories. We integrated BLIP3 for image captioning, achieving an average CLIPScore of 0.2852, and utilized advanced LLMs for product link retrieval. While our implementation of LLaMA3 achieved up to 0.97 accuracy in matching products to their purchase sources, ablation studies revealed superior product link retrieval performance when bypassing BLIP3 and OCR integration. This suggests that advanced LLMs can effectively handle multimodal tasks directly. Through our system, we demonstrate a practical application for monetizing advanced AI models by creating value in the e-commerce ecosystem, transforming passive content consumption into revenue-generating opportunities through contextually relevant, non-intrusive product discovery.**

Keywords—Interactive Advertising, Object Detection, Optical Character Recognition (OCR), Image Captioning, LLMs

I. Introduction

Interactive advertising presents a promising avenue for monetizing Large Language Models (LLMs), whose training costs have grown dramatically at a rate of 2.4× per year since 2016, with frontier models projected to cost over a billion dollars by 2027 [4]. This monetization need coincides with significant opportunities in AI-driven marketing, where the market was estimated at $15.84 billion in 2021 and is predicted to reach $107.5 billion by 2028 [6].Through the combination of computer vision and LLMs, this work proposes a system that enables seamless product discovery and purchasing within visual media.

Current solutions have notable limitations. Google Lens enables users to search for products in images (Gour et al., 2024) but requires active user initiation through photos or screenshots, and often returns general information rather than direct purchase options. Meanwhile, Kerv Interactive's technology [7] requires pre-embedding interactive elements into videos before deployment, limiting scalability and real-time applications.Recent advances in computer vision and natural language processing provide new opportunities to address these limitations. Traditional object detection methods have evolved toward CNN-based approaches like YOLO frameworks, with YOLOv8 particularly suited for real-time applications through its enhanced speed-accuracy trade-offs and advanced feature extraction [9]. For generating rich image descriptions, transformer-based architectures like BLIP3 have demonstrated superior performance through multimodal encoder-decoder designs and extensive pre-training [2]. Additionally, recent work has shown how Multimodal Large Language Models can effectively handle tasks ranging from contextual object detection to product retrieval, while maintaining high accuracy and scalability [10].

Our project addresses existing limitations by combining cutting-edge models into an automated pipeline. The system leverages an ensemble object detection approach, combining a fine-tuned YOLOv8 model trained on 25 custom product classes (achieving mAP@50 of 0.535) with the base YOLOv8 model's 64 product classes, enabling detection across 89 consumer product categories. This pipeline integrates BLIP3 for image captioning, achieving an average CLIPScore of 0.2852, and utilizes advanced LLMs for product link retrieval, with LLaMA3 up to 0.97 accuracy in matching products to their purchase sources. Notably, our ablation studies revealed that product link retrieval accuracy improved without BLIP3 and OCR integration, suggesting that advanced LLMs can effectively handle multimodal tasks directly.

The InvadeAI framework automates traditionally manual processes like product tagging and link creation while maintaining high accuracy. This approach transforms passive content consumption into revenue-generating opportunities through contextually relevant, non-intrusive product discovery. By enabling automatic identification and seamless purchasing of products within visual content, our system demonstrates a practical application for monetizing advanced AI models while creating value in the e-commerce ecosystem.

II. Dataset Description and Preprocessing

We used the Open Images Dataset V7, a large-scale dataset featuring 1.9 million images and 16 million annotated bounding boxes spanning 600 classes. From its 1.7 million training images, approximately 71.9% include at least one product-related object, and among these product categories, clothing, footwear, and windows appear frequently. To select classes suitable for our YOLOv8 model fine-tuning, we

† Equal First Authors
* Mohammad Masum is the Corresponding Author

2025 IEEE Conference on Artificial Intelligence (CAI) | 979-8-3315-2400-5/25/$31.00 ©2025 IEEE | DOI: 10.1109/CAI64502.2025.00027

979-8-3315-2400-5/25/$31.00 ©2025 IEEE
DOI 10.1109/CAI64502.2025.00027

Authorized licensed use limited to: California State University - EAST BAY. Downloaded on August 29,2025 at 07:47:04 UTC from IEEE Xplore. Restrictions apply.

prioritized specific, commercially relevant categories absent from the original YOLOv8 classes, avoiding broad categories like "Clothing" or "Electronics" due to their significant internal visual diversity. We ultimately chose 25 classes demonstrating consistent visual characteristics, such as "Footwear" and "Camera," ensuring reliable model convergence during transfer learning [1]. Each class has at least 4,500 annotated images (4,000 training, 500 validation), determined as an optimal balance between model performance and computational efficiency.

A. Image Preprocessing

Figure 1 shows a sample image with ground truth bounding boxes from the Open Images dataset. A filtered dataset was then created, containing only the selected product classes for training in the detection model. The image and label organization follows the YOLO framework requirements, with separate directories for training and validation sets, each containing corresponding image and label subdirectories. Label preprocessing involved the conversion of Open Images format annotations to YOLO format, where bounding box coordinates were normalized to the 0-1 range. Individual label files were generated for each image, containing the class index and normalized bounding box coordinates in the format: *class index, x center, y center, width, and height*. Validation was performed to ensure all bounding boxes fell within valid ranges.

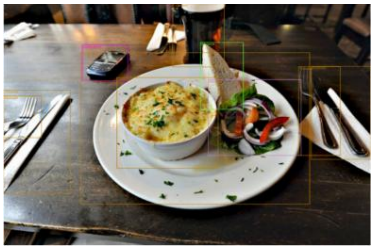


Figure 1: Example image with ground truth bounding boxes from the Open Images Dataset

Data validation was implemented through multiple checks, including verification of matched image-label pairs, confirmation of minimum image requirements per class, and validation of label format and class indices. A dataset.yaml configuration file was generated to specify the training and validation image directories, number of classes, and class names, with subsequent validation performed using YOLO's built-in verification systems. Additionally, the BLIP3 and OCR output text underwent preprocessing to enhance LLM prompt effectiveness. This process included the removal of special characters and extra whitespace, along with standardization of text case and formatting. The processed text was then combined with the cropped object images to provide prompt input to the LLM.

III. Methodology

A. Proposed Framework

We propose a novel framework Invade AI for product link retrieval, as illustrated in Fig. 2. The framework is implemented with a user interface centered around a YouTube video player window augmented with an interactive product detection overlay, as shown in Fig. 3. The input video stream is processed by an ensemble detection model comprising two components: a base YOLOv8 model that can identify 64 product-related classes, and a finetuned YOLOv8 model specifically trained on an additional 25 product classes not present in the base model. When the base YOLOv8 model identifies objects in the image, its detections undergo a filtering process to retain only the 64 product-related classes from its total class set, eliminating irrelevant classes such as people or animals. These filtered detections are then combined with the outputs from the fine-tuned YOLOv8 model, which contributes detections from its 25 specialized product categories. This ensemble approach expands the system's product recognition capabilities to 89 distinct product classes.

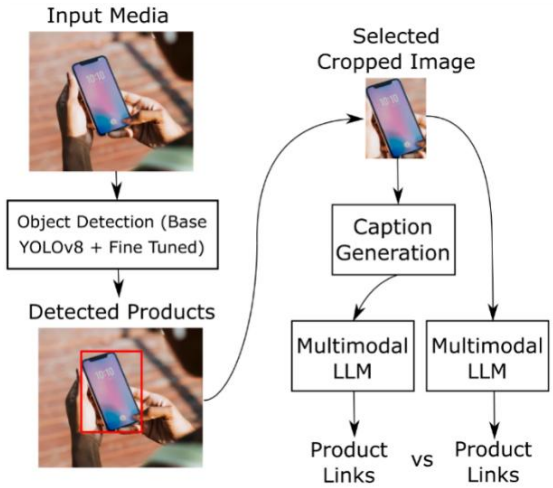


Figure 2: Our proposed framework InvadeAI - integrating YOLOv8 for object detection, BLIP3 for caption generation, and an LLM to dynamically enhance product link retrieval.

The interface preserves a clean viewing experience by only displaying product bounding boxes when the video is paused, with an optional "show all" mode that displays all detections versus only showing them when hovering over a specific box. The UI implements event handling to ensure proper interaction with both video controls and detected products, such that the object detection elements don't interfere with pausing and playing of the video player.

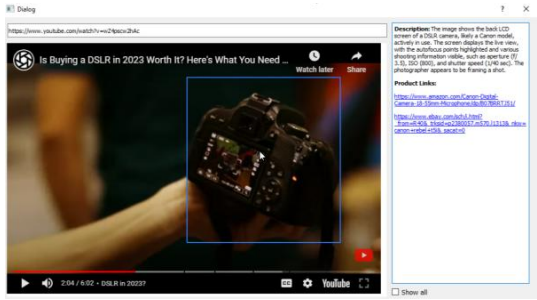


Figure 3: Example of an interactive advertising system within a user interface. When a user hovers over a product in the video, its bounding box appears. Clicking the box prompts the system to retrieve and display the corresponding product links in the UI. Video

When a user expresses interest in a specific product by clicking on it in the interface, the system utilizes the corresponding bounding box coordinates to extract a cropped image of the selected product. This cropped image then flows through two processing streams: BLIP3 caption generation and optical character recognition (OCR). BLIP3 generates a detailed description of the product's visual characteristics, while OCR extracts any visible text present in the product image, such as brand names or product information.

Authorized licensed use limited to: California State University - EAST BAY. Downloaded on August 29,2025 at 07:47:04 UTC from IEEE Xplore. Restrictions apply.

The final stage of the pipeline involves a multimodal LLM that receives a prompt constructed from four elements: the cropped product image, the BLIP3-generated caption, any text extracted through OCR, and a query asking the model to find links to purchase products related to the input. The output links are then presented to the user through the interface, completing the interactive advertisement experience. This end-to-end system creates a bridge between video content and e-commerce functionality, enabling natural product discovery without disrupting the viewing experience. Additionally, two versions of the system are developed and tested by implementing the pipeline both with and without the generated captions from BLIP3 and OCR. Given the high performance of advanced LLMs, it is possible equivalent performance can be achieved without the additional caption generation step, making for a more lightweight system.

B. Object Detection

The object detection system employs an ensemble approach combining a base YOLOv8s model with a fine-tuned counterpart. YOLOv8, introduced by Ultralytics [8] in 2023, represents an advancement in real-time object detection, featuring architectural improvements such as a CSPNet backbone, FPN+PAN neck structure, and an anchor-free detection approach. The model architecture, consists of several components that work in tandem to achieve efficient object detection. The backbone begins with an input image of size 640x640x3, followed by a series of convolutional layers and feature aggregation modules that progressively extract and refine visual features [8]. The base YOLOv8 model's capabilities were extended through transfer learning on the final output layer to accommodate an additional 25 product-related classes. This was accomplished by fine-tuning the model's detection head while preserving the pre-trained backbone and neck weights.

C. Caption Generation and Product Link Retrieval

Caption generation is essential for building an effective interactive advertising system to generate rich and contextually relevant descriptions for visual content. This project combines BLIP3 for image captioning and Optical Character Recognition (OCR) to extract text, creating a complete understanding of visual and textual elements. These technologies generate detailed captions and prompts, enabling product link retrieval through LLMs.

Images are preprocessed before applying BLIP3 and OCR to ensure they meet the compatibility and requirements of BLIP3's multimodal encoder-decoder architecture and OCR. This preprocessing involves detecting and cropping objects of interest using YOLOv8 to focus on relevant image regions. The cropped images are then resized and normalized to align with BLIP3's input specifications. Additionally, OCR extracts textual elements from the images, such as labels, brand names, or embedded product details, enhancing the visual content's overall understanding and contextual richness.BLIP3 employs a transformer-based multimodal encoder-decoder architecture which has been pretrained on enough image-text datasets such as BLIP3-KALE [2]. It generates captions that are both semantically accurate and contextually rich. In this paper, we applied BLIP3, including its variant from the advanced xGen-MM series, which is based on the Salesforce/xgen-mm-phi3-mini-instruct-r-v1, building on the success of the blip series. In the caption generation process, cropped images from YOLOv8 object detection are processed through the BLIP3 encoder, which extracts detailed visual features embedded in a multimodal space. These features are then passed to the decoder, which leverages its pre-trained linguistic knowledge to produce descriptive and contextually relevant captions.

Optical Character Recognition (OCR) uses tools such as Python-tesseract (pytesseract) to identify and extracts textual elements like brand names, product labels, and specifications from the cropped regions. The extracted text is cleaned to remove any errors or inconsistencies, ensuring the final output is clear and accurate. Finally, the captions generated by BLIP 3 and OCR outputs for the cropped image are combined to form a structured prompt for the LLM tasked with product link retrieval. The quality of the captions generated by BLIP3 is evaluated using CLIPScore, a metric designed to measure the semantic alignment between a caption and its corresponding image. CLIPScore utilizes a pre-trained CLIP model to compute the similarity between the embeddings of the image and the caption [5]. The process begins by inputting the cropped image and its generated caption into the CLIP model, which extracts embeddings for both modalities. A cosine similarity measure is then calculated between the normalized embeddings, providing a quantitative score. The average CLIPScore is calculated across all evaluated captions for providing an overall measure of the system's performance, indicating the consistency and reliability of the generated descriptions. CLIPScore confirm that the captions are accurate and relevant to the images, making them useful for tasks like product identification and interactive advertising. Finally for the target product images, both LLaMA3 and Gemini 1.5 Pro were utilized to provide informative descriptions and obtain links to make purchases.

IV. Experimental Design

We fine-tuned YOLOv8 using the following hyperparameters: epochs=300, image size=640, batch size=16, workers=6, patience=20, and optimizer=Adam. For implementation, we employed ultralytics for YOLOv8 training and inference, PyTorch for deep learning, OpenCV-python for image processing, NumPy and Pandas for numerical computations and data manipulation, matplotlib for visualization, transformers for the BLIP3 model, pytesseract for OCR, and groq for LLM API integration. The experiments were conducted using a Windows 10 PC (AMD Ryzen 3900X CPU, 128GB RAM, NVIDIA GeForce RTX3080 GPU, 2TB SSD) and the Groq API for cloud computing. The system utilizes YOLOv8, BLIP3, OCR, and LLM models for inference tasks.

A. Training Data Scaling Experiment

We conducted a scaling study, to optimize the training dataset composition for the fine-tuned YOLOv8 model to determine whether balancing training data by image count or bounding box count would yield better results and to identify the optimal number of samples per class, focusing on two representative product classes: Hat and Tire. Multiple object detection models were trained independently for each class, varying the number of training images from 100 to 5500. We evaluated models' performance using mAP@50, with performance curves plotted against both image count and bounding box count.

B. Prompt Engineering

We used a structured prompt template for product link retrieval. The captions generated from the BLIP3 output are

Authorized licensed use limited to: California State University - EAST BAY. Downloaded on August 29,2025 at 07:47:04 UTC from IEEE Xplore. Restrictions apply.

used as the input for the prompt to the LLM. The LLM then retrieves the product links from the brand websites and ecommerce stores for the products in the cropped image. This template integrates the cropped object image, OCR text, and BLIP3 caption information in a format designed to elicit direct product purchase links from the LLM. The prompt structure maintains consistency while preserving context from multiple input sources, providing a direct link to purchase this or similar products. This approach ensures focused and relevant responses from the LLM while maintaining the contextual richness of the multi-modal input.

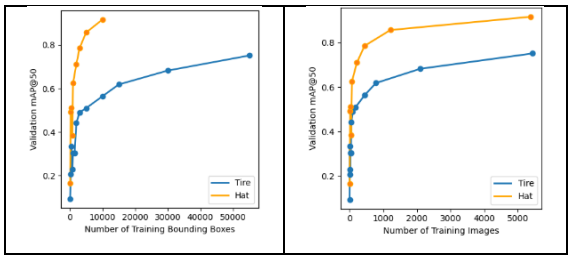


Figure 4: Performance scaling comparison of single-class models for the Hat and Tire classes. Plots show mAP@50 versus training dataset size as a function of the number of bounding boxes (top) and the number of training images (bottom).

Table 1: Summary of the various LLMs

Parameter	Llama 3.2	Gemini 1.5 pro	BLIP3
Model size (parameters)	$9x10^{10}$	-	$1.51x10^{6}$
Temperature	0.5	1.0	-
Top p	0.9	0.95	-
Max tokens	2048	8192	-

As shown in Table I, both LLaMA 3.2 (90 billion parameters) and Gemini 1.5 Pro were utilized with their default hyperparameters for link retrieval tasks. LLaMA 3.2 used its default configuration with a temperature of 0.5 and top_p of 0.9, while Gemini 1.5 Pro used its standard default values with a temperature of 1.0 and top_p of 0.95. These models interface with the Groq API to retrieve links from relevant websites, with default maximum token lengths of 2048 for LLaMA 3.2 and 8192 for Gemini 1.5 Pro.

C. Model Evaluation

We randomly selected 200 images per class from the Open Images test dataset to create a balanced test set across various product categories. The primary evaluation metric was mean Average Precision at 50% Intersection over Union (mAP@50), which provided an overall measure of detection accuracy. Additionally, we computed precision, recall, and F1-score at the confidence threshold that maximized the F1-score. These metrics were evaluated both in aggregate across all classes and on a per-class basis, facilitating a detailed analysis of model performance across different product categories. To assess the quality of captions generated by BLIP3, we employed CLIPScore, which measures the semantic similarity between the generated captions and their corresponding images.

The performance of product link retrieval by the LLMs was evaluated through manual assessment under two different prompting conditions. In the first condition, LLMs were provided with the complete input: cropped product image, BLIP3-generated caption, OCR-extracted text, and the retrieval query. In the second condition, only the cropped image and query were provided, allowing assessment of the impact of additional textual context. Accuracy was calculated as the ratio of correct product links to total retrieved links. A product link was considered correct if it led to a product matching the detected object in terms of category. For the prompting condition without BLIP-3, the LLM could fetch the accurate product links each time an input image was given.

V. Experimental Results

A. YOLOv8 Results

The training scaling experiment revealed that balancing the number of images, rather than bounding boxes, across classes was more appropriate for the multi-class model. As shown in Fig. 4, single-class models for both Hat and Tire reach performance plateaus at approximately 4000 training images. However, examining the performance against the number of bounding boxes reveals a significant disparity: the Tire class requires substantially more bounding boxes to reach its performance plateau compared to the Hat class. This disparity arises from images containing Tire typically having multiple instances per image, while Hat images usually contain fewer instances. Despite this difference in bounding box density, both classes show similar performance scaling with respect to the number of training images, supporting the decision to balance the dataset based on images per class rather than total bounding boxes.

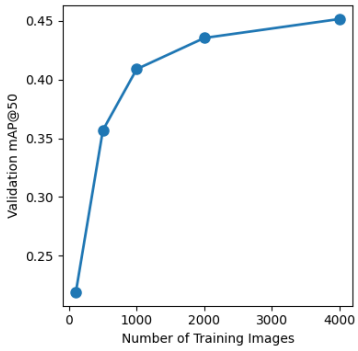


Figure 5: multi-class model performance (mAP@50)

The scaling behavior observed in single-class experiments was further validated in the multi-class setting. As shown in Fig. 5, the multi-class model's performance follows a similar pattern, with mAP@50 improvements diminishing beyond 4000 images per class. This consistent behavior across both single-class and multi-class scenarios confirms the appropriateness of the 4000 images per class allocation for the final training dataset. For optimal model deployment, class-specific confidence threshold calibration was performed using F1 score optimization. Fig. 6 illustrates this process for the Suit class, where precision, recall, F1 score, and IoU metrics are plotted against confidence threshold values. This calibration was conducted for each class using the test dataset.

The final model achieved an overall mAP@50 of 0.535 across all classes, as detailed in Table 2. Performance varied considerably across classes, with 21 classes achieving mAP@50 of 0.45 or higher, indicating robust detection capabilities for most categories. However, certain classes like Fashion accessory (mAP@50 = 0.130) and Musical instrument (mAP@50 = 0.114) showed notably poor performance, suggesting these categories may be too broad or diverse for effective learning.

Authorized licensed use limited to: California State University - EAST BAY. Downloaded on August 29,2025 at 07:47:04 UTC from IEEE Xplore. Restrictions apply.

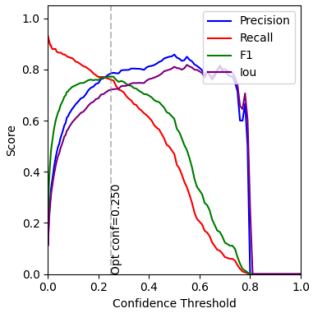


Figure 6: Confidence threshold calibration for the Suit class.

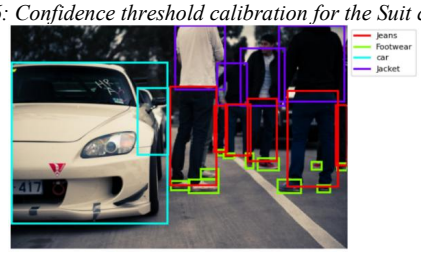


Figure 7: Sample product detection results from the YOLOv8

The optimal confidence thresholds also varied widely across classes, ranging from 0.022 to 0.580. This variation suggests that implementing class-specific confidence thresholds through a lookup table could significantly improve detection reliability and user experience. The fine-tuned model was integrated with the base YOLOv8 model to create an ensemble system capable of detecting 89 product classes, combining the 25 newly trained classes with the 64 product-related classes from the base model. Fig. 7 demonstrates this expanded detection capability, showing the ensemble model identifying multiple product categories from both the fine-tuned and base YOLOv8 models.

B. Caption Generation Results

The caption generation results were evaluated using CLIPScore, which evaluates the alignment between the generated captions and the corresponding image. Table 3 illustrates some sample iamges, corresponding generated captions, and associated CLIPScores. The average CLIPScore across all evaluated images is 0.2852.

C. Product Link Retrieval Results

The accuracy of the product links retrieved by the LLMs was assessed, with particular attention given to the LLM outputs for input prompts both with and without captions generated from BLIP3 and OCR. Despite the absence of text in some photos, LLaMA3 and Gemini 1.5 Pro exhibited remarkable skills in generating captions. For example, LLaMA3 accurately retrieved links to manufacturer pages and successfully identified the make and model of cars in images, such as the Hyundai Elantra and Toyota Corolla, based solely on visual clues. Meanwhile, Gemini 1.5 Pro collected links from well-known ecommerce sites like Amazon and provided detailed descriptions for appliances and gadgets, correctly identifying products such as "front-loading washing machines" and "flat-screen televisions".

D. Ablation Studies

An ablation study was conducted by evaluating the performance of LLaMA3 and Gemini in generating captions and retrieving product links without relying on BLIP3 and OCR (Table 4). The ablation study analyzes the impact of removing BLIP3 for caption generation. The results indicate that even without BLIP3 and OCR, LLaMA3 and Gemini can effectively generate captions and retrieve product links from images.

Table 2: Summary of per-class performance metrics for the YOLOv8 fine-tuned model.

Class	mAP @50	opt conf	f1	precision	recall	mean IoU
All Classes	0.535	0.130	0.549	0.514	0.589	0.464
Suit	0.651	0.250	0.751	0.735	0.767	0.681
Glasses	0.676	0.065	0.785	0.860	0.722	0.699
Dress	0.463	0.100	0.593	0.670	0.532	0.610
Jeans	0.493	0.130	0.633	0.640	0.626	0.583
Tire	0.596	0.200	0.626	0.764	0.530	0.662
Fashion Accessory	0.130	0.085	0.345	0.553	0.251	0.477
Microphone	0.479	0.180	0.570	0.682	0.490	0.614
Guitar	0.873	0.180	0.859	0.843	0.876	0.753
Toy	0.397	0.260	0.445	0.411	0.485	0.394
Poster	0.694	0.360	0.742	0.839	0.664	0.799
Drink	0.287	0.130	0.397	0.393	0.402	0.361
Bicycle wheel	0.776	0.290	0.778	0.889	0.692	0.793
Furniture	0.070	0.022	0.183	0.145	0.248	0.165
Door	0.564	0.330	0.573	0.636	0.522	0.574
Jacket	0.602	0.110	0.726	0.770	0.688	0.685
Baked goods	0.492	0.580	0.537	0.540	0.535	0.546
Desk	0.464	0.210	0.583	0.573	0.594	0.534
Hat	0.609	0.170	0.805	0.834	0.778	0.761
Flag	0.690	0.410	0.750	0.757	0.743	0.687
Shorts	0.598	0.330	0.714	0.729	0.699	0.634
Musical instrument	0.114	0.150	0.209	0.184	0.242	0.211
Shirt	0.562	0.110	0.723	0.729	0.717	0.676
Camera	0.834	0.440	0.819	0.789	0.851	0.732
Beer	0.777	0.430	0.801	0.836	0.769	0.758
Footwear	0.497	0.085	0.618	0.719	0.541	0.594

Table 3: Examples of generated captions for cropped images processed by the BLIP3 model

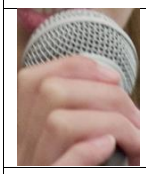
Image	Generated Caption	CLIPS core
	a woman holding a microphone in her hand	0.3369
	a cake with purple and yellow decorations	0.3941
	a bottle of beer sitting on top of a table	0.2912

Table 4: Example of retrieved product links from LLMs for a given product class

Product Class	Model	Product Links
Multiple Electronics	Gemini 1.5 pro	https://www.amazon.com/s?k=refrigerator&i=appliances https://www.amazon.com/s?k=washing+machine&i=appliances
Multiple Electronics	Llama 3	https://www.lg.com/ https://www.sony.com/

Authorized licensed use limited to: California State University - EAST BAY. Downloaded on August 29,2025 at 07:47:04 UTC from IEEE Xplore. Restrictions apply.

Product Class	Model	Product Links
Cars	Gemini 1.5 pro	https://www.whirlpool.com/ https://www.toyota.com/corolla/ https://www.hyundaiusa.com/us/en/vehicles/elantra
Cars	Llama 3	https://www.hyundaiusa.com/ https://www.toyota.com/
Smartphone	Gemini 1.5 pro	https://www.oneplus.com/oneplus-11 https://store.google.com/product/pixel7 pro https://www.apple.com/iphone-14-pro/
Smartphone	Llama 3	https://www.oneplus.com/ https://www.apple.com/ https://store.google.com/

Table 5 demonstrate that removing BLIP3 and OCR from the system for LLaMA3 does not negatively affect the ability to retrieve product links. In fact, the product link accuracy was actually improved without the use of BLIP3 and OCR. This is particularly evident in the case of the Clothes class with Gemini 1.5 Pro, where accuracy improved from 82% with BLIP3 and OCR to 95% without. This suggests that the additional text from BLIP3 and OCR is potentially distracting to the LLMs in finding product links for a target image.

Table 5: LLM product link retrieval accuracy with and without input captions

Model	Accuracy with BLIP3 and OCR	Accuracy without BLIP3 and OCR
LLaMA3	83%	91%
Gemini 1.5 Pro	85%	93%
LLaMA3	81%	92%
Gemini 1.5 Pro	82%	95%
LLaMA3	90%	97%
Gemini 1.5 Pro	94%	86%

VI. Limitations and Future Works

While our results demonstrate promising technical feasibility, performance varies significantly across categories; notably, detection accuracy is low for diverse classes like Fashion accessory (mAP@50=0.130). Additionally, although BLIP3 produces contextually relevant captions (CLIPScore=0.285), caption quality remains inconsistent. Our manual verification of product links could benefit from automated evaluation methods to ensure accuracy rigorously. We acknowledge this study did not extensively cover user-experience factors, real-world integration, or automated link verification—areas crucial for practical deployments. Future work should explore UI improvements, integration into e-commerce platforms, automated validation methods, and leveraging lightweight models like LLaMA-derived architectures (LLaMA 11B, LLaVA 7B) to facilitate efficient, scalable deployments.

VII. Conclusion

Through the integration of large language models (LLMs) and computer vision, this research created a novel interactive framework InvadeAI for advertising system. The framework provides a smooth pipeline that connects visual content with e-commerce by utilizing LLaMA3 and Gemini 1.5 Pro for natural language processing and product link retrieval, and YOLOv8 for accurate object detection. Notably, while BLIP3 and OCR were integrated for caption generation and text extraction, ablation studies revealed that the LLMs actually outperformed at product link retrieval without these additional components. This finding suggests that advanced LLMs can effectively handle multimodal tasks directly, potentially simplifying the system architecture while improving performance. The successful implementation of this framework demonstrates a viable path toward monetizing costly LLM infrastructure through interactive advertising, offering a practical solution that benefits both media providers and end users while transforming traditional advertising approaches.

References

[1] A. Torralba, K. P. Murphy, and W. T. Freeman, "Sharing visual features for multiclass and multiview object detection," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 28, no. 8, pp. 1-17, 2006.

[2] Awadalla, A., Xue, L., Shu, M., Yan, A., Wang, J., Purushwalkam, S., Shen, S., Lee, H., Lo, O., Park, J. S., Guha, E., Savarese, S., Schmidt, L., Choi, Y., Xiong, C., & Xu, R. (2024). BLIP3-KALE: Knowledge Augmented Large-Scale Dense Captions. ArXiv.org. https://arxiv.org/abs/2411.07461

[3] Camera & Lens Compare, "Is Buying a DSLR in 2023 Worth It? Here's What You Need to Know!" Youtube, 2023. [Online]. Available: https://www.youtube.com/watch?v=wZ4pscw2hAc

[4] Cottier, B., Rahman, R., Fattorini, L., Maslej, N., & Owen, D. (2024). The Rising Costs of Training Frontier AI Models. arXiv preprint arXiv:2405.21015.

[5] J. Hessel, A. Holtzman, M. Forbes, R. Le Bras, and Y. Choi, "CLIPScore: A Reference-free Evaluation Metric for Image Captioning," arXiv preprint, 2021.

[6] Kshetri, N., Dwivedi, Y. K., Davenport, T. H., Panteli, N. (2024). Generative Artificial Intelligence in Marketing: Applications, Opportunities, Challenges, and Research Agenda. International Journal of Information Management, 74(2), 102700.0

[7] Pate, M. T., Roque, M., Welch, A. (2023). Connected interactive content data creation, organization, distribution and analysis. U.S. Patent No. 11,617,015. Washington, DC: U.S. Patent and Trademark Office.

[8] Ultralytics, "YOLOv8 anchor-free bounding box prediction - issue 189," GitHub repository, 2023. [Online]. Available: https://github.com/ultralytics/ultralytics/issues/189.

[9] Yaseen, M. (2024). What is YOLOv9: An In-Depth Exploration of the Internal Features of the Next-Generation Object Detector. ArXiv.org. https://arxiv.org/abs/2409.07813

[10] Zang, Y., Li, W., Han, J., Zhou, K., & Chen Change Loy. (2024). Contextual Object Detection with Multimodal Large Language Models. International Journal of Computer Vision. https://doi.org/10.1007/s11263- 024-02214-4

Authorized licensed use limited to: California State University - EAST BAY. Downloaded on August 29,2025 at 07:47:04 UTC from IEEE Xplore. Restrictions apply.